Chapter 5 Data compression

Ch 5.1 – 5.6

C: Information / Data → Code ·, –, ·–, –·· 4 syb

Morse code: Alphabet → seq of symb

$e \to C(e) = \cdot$

$a \to C(a) = \cdot -$

most freq letter → shortest
least " → longest.

$\underline{e\ i\ t\ a\ n}$ ····· $\underline{p\ x\ z\ q\ y}$

short code   long code

Object: given X: r.v.'s w/ p.m.f $p(x)$

§5.1 Examples of codes

D^*: sequence of D-ary alphabet

$D = \{0, \cdots, D-1\}$

2-ary $D = \{0, 1\}$

Def $\boxed{\text{Source code}}$ C

$C: \mathcal{X} \to D^*$

$x \in \mathcal{X}$, $C(x) \in D^*$: $C(x) = 0, 1, 00, 11$

$\boxed{l(x): \text{length of } C(x)}$

Ex $\mathcal{X} = \{\text{red}, \text{blue}\}$

$C(\underline{\text{red}}) = \underline{00}$ $C(\underline{\text{blue}}) = \underline{11}$

$l(\text{red}) = 2$ $l(\text{blue}) = 2$

Def $\boxed{\text{Expected length } L(C)}$ of C.

for random X. p.m.f $P(x)$

$$L(C) = \sum_{x \in \mathcal{X}} P(x) l(x)$$

Ex 5.1.1 X: rand var with

	$Pr(X=x)$	$C(x)$	$l(x)$
1	$\frac{1}{2}$	0	1
2	$\frac{1}{4}$	10	2
3	$\frac{1}{8}$	110	3
4	$\frac{1}{8}$	111	3

$H(X) = 1.75$

$L(C) = 1.75$.

$\underbrace{0}_{1}\,\underbrace{110}_{3}\,\underbrace{111}_{4}\,\underbrace{10}_{2}\,\underbrace{0}_{1}\,\underbrace{110}_{3}$

1. $H(X) = \underline{L(C)}$

2. Uniquely decodable,

Ex 5.1.2

x	Pr	$C(x)$	$l(x)$
1	$\frac{1}{3}$	0	1
2	$\frac{1}{3}$	10	2
3	$\frac{1}{3}$	11	2

$H(X) = 1.58$

$L(C) = 1.666$

1 $H(X) < L(C)$

2. Unique decodable

5.2

$H(X) \underline{\leq L(C)}$

Def A code is $\boxed{\text{nonsingular}}$

if $\forall x, x'$

$$x \neq x' \Rightarrow C(x) \neq C(x')$$

(1-1 mapping)

$C(x) = 00$

$C(x') = 01$

$x x' \cdots x_n \rightarrow C(x) \; C(x') \; C(x_n)$

Def The extension C^* of a code C

$$C(x_1 \cdots x_n) = C(x_1) \cdots C(x_n)$$

Ex $C(x_1) = 00$, $C(x_2) = 11$ $\Rightarrow C(x_1 x_2) = 0011$

Def A code is $\boxed{\text{Uniquely decodable}}$

If its extension is nonsingular

$x_1 \cdots x_n$: 1-1 mapping

$(11010 \cdots\cdots 10 \cdots 1000) \rightarrow$

$\rightarrow$

Instantaneously vs read all

Def A code is a $\boxed{\text{prefix code}}$

or an $\boxed{\text{instantaneous code}}$

If no codeword is a prefix of any codeword.

Ex) $C(x_1) = \underline{11}$
$C(x_2) = \underline{110}$. not prefix

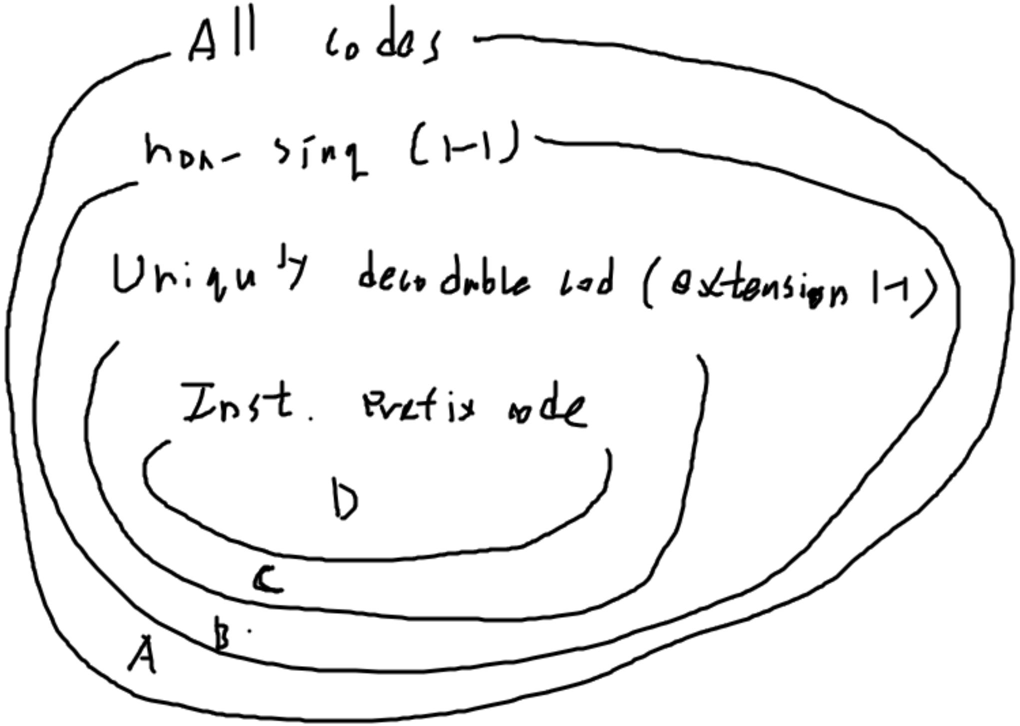


Table 5-1

X	A	B	C	D
	non singular			
1	0	0	10	0
2	0	010	00	10
3	0	01	11	110
4	0	10	110	111

C ⟺ D

prefix

1-1. (1100 ......)

3 2 ...

010 → 2, 14, 31

1100 → 4 2

1100... → 3 1

1 What is the most efficient (shortest) int. code → §5-2 → 5.4

2 How can we construct inst. code §5-6.

§ 5.2 Kraft ineq

(lower bound of inst. code)

Thm 5.2.1. For inst. code over an alphabet size D.

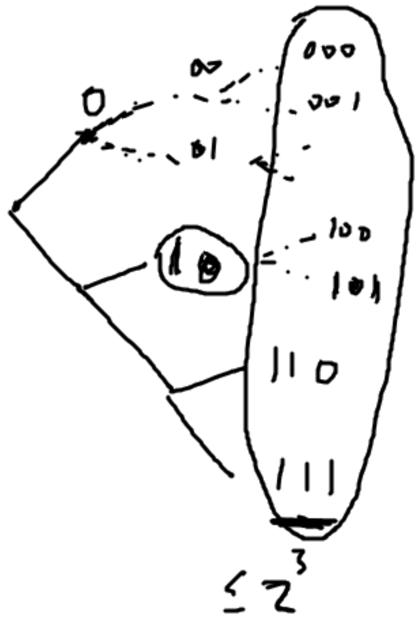
code word length $l_1, l_2, \dots, l_m$ must satisfy ($l_1 = 1$, $l_2 = 2$)

$$\boxed{\text{kraft}} \quad \sum_{i=1}^{m} D^{-l_i} \le 1 . \quad D^{-l_1} + \dots + D^{-l_m} \le 1$$

Converse. If a set of code word length satisfy kraft ineq

$\Rightarrow \exists$ inst code.

Pf) 2-ary tree Fig 5.2



\# eliminate

0's desc	2^{3-1}
10's des	2^{2-1}
110 →	2^{3-3}
111 →	2^{3-3}
+	

$\le 2^3$

$$\Rightarrow \sum_{i=1}^{m} 2^{l_{max} - l_i} \le 2^{l_{max}}$$

$$\Rightarrow \sum_{i=1}^{m} D^{l_{max} - l_i} \le D^{l_{max}}$$

$$\Rightarrow \sum_{i=1}^{m} D^{-l_i} \le 1.$$

Thm 5.2.2 (Extended Kraft ineq)

For any countably infinite set of codewords of inst. code

$$\sum_{i\geq 1}^{\infty} D^{-l_i} \leq 1$$

converse : true.

Pf) we 2-ary alphabet., y_i : code word.

Let $0.y_1 \cdots y_{l_i} = \sum_{j\geq 1}^{l_i} y_j D^{-l_j}$ — 0, 1

we define

$$\left[0.y_1 \cdots y_{l_i},\ 0.y_1 \cdots y_{l_i} + \frac{1}{D^{l_i}}\right)$$

For example

$C(y_1) = \underset{2}{\underbrace{10}}$

$0.10 = 10 \cdot 2^{-2} = 10 \cdot 0.01$

$= 0.10$

$\left[0.10,\ 0.10 + \frac{1}{2^2}\right)$

$\frac{1}{2^2} = 0.01$

$[0.10,\ 0.11)$ 101 1001
100 1000.

: includes all code word begin with 10

$\bigcup [\ ) \subset [0,1)$

$$\sum_{i\geq 1}^{\infty} D^{-l_i} \leq 1$$

□

§ 5.3 Optimal code

Object : Finding $\underline{\text{inst. code}}$ with minimum $E\,l(X)$

$l_i \in \mathbb{R}$.

$$\boxed{\begin{array}{l} \text{Minimize } L = \sum p_i l_i \\ \text{satisfying } \sum D^{-l_i} = 1 \end{array}}$$

Lagrange multiplier

$$J = \sum p_i l_i + \underline{\lambda} \left(\sum D^{-l_i} \right)$$

$$\Rightarrow \frac{\partial J}{\partial \underline{l_i}} = p_i - \lambda D^{-l_i} \log_e D = 0$$

$$\underline{D^{-l_i} = \frac{p_i}{\lambda \log_e D}}$$

$$1 = \sum D^{-l_i} = \sum \frac{p_i}{\lambda \log_e D} = \frac{1}{\underset{\sim}{\lambda} \log_e D} =$$

$$\lambda = \frac{1}{\log_e D}$$

Optimal length $\Rightarrow$ $\underline{l_i^*} = -\log_D p_i$

Expected length $\Rightarrow$ $L^* = \sum p_i l_i^*$

$$= -\sum p_i \log_D p_i$$

$$:= H_{\underline{D}}(X)$$

Thm 5.3.1

The expected length L of $\boxed{\text{inst}}$ D-ary code satisfy

$$L \geq H_D(X)$$

Equality holds when $D^{-l_i} = p_i$

Pf) $L - H_D(X) = \sum p_i l_i - \sum p_i \log_D p_i$

$$= -\sum p_i \log_D D^{-l_i} - \sum p_i \log_D p_i$$

$$\pm \; p_i \log_D \left(\sum_j D^{-l_j} \right)$$

we define $r_i = \frac{D^{-l_i}}{\sum_j D^{-l_j}}$ $= 1$

$D(p \| r) \geq 0$

$$L - H_D(X) = \underbrace{\sum p_i \log_D \left(\frac{p_i}{r_i} \right)}_{}$$

≤ 1

$$\underbrace{-\sum p_i \log_D \left(\sum D^{-l_i} \right)}_{\geq 0} \geq 0$$

Equality $D(p \| r) = 0$ $\quad \frac{p_i}{r_i} = 1$

$\sum D^{-l_i} = 1$

$$\frac{p_i}{r_i} = \frac{p_i}{D^{-l_i}} = 1$$

$p_i = D^{-l_i}$ $\quad$ $p_i = D^{-n}$

$\Longleftrightarrow$ $l_i = -\log_D p_i$ $\quad$ $\boxed{//\!/}$

Def A probility dis is called D-adic if each prob is equal to D^{-n} for some n.

X : D-adic $\Rightarrow$ $-\log_D p_i$: integer

$l_i^* = n$.

§ 5.4 Suboptimal procedur (Shannon-Fano)

§ 5.6 optimal " (Huffman)

§ 5.4 Bounds on the optimal code length

$l_i = -\log_D p_i \neq$ integer

$$l_i = \left\lceil \log_D \frac{1}{p_i} \right\rceil$$

$\lceil x \rceil$: smallest integer $\geq x$

$$\log_D \frac{1}{p_i} \leq \left\lceil \log_D \frac{1}{p_i} \right\rceil$$

$$\Rightarrow \underline{\sum_i D^{-\lceil \log_D \frac{1}{p_i} \rceil}} \leq \underline{\sum_i D^{-\log_D \frac{1}{p_i}}} = \underline{1}$$

$$\log_D \frac{1}{p_i} \le l_i = \left\lceil \log_D \frac{1}{p_i} \right\rceil < \log_D \frac{1}{p_i} + 1$$

$\sum_i p_i$

1 bit.

$$\Rightarrow H_D(X) \le L < H_D(X) + 1$$

$l_1^* \cdots l^*$

Thm 5.4.1 Let $l_1^*, \cdots l_m^*$ optimal code length.

Then $H_D(X) \le L^* < H_D(X) + 1$

Pf) $l_i^* \le l_i$

$L^* \le L$

$H_D(X) \le L^* \le L < H_D(X) + 1$ $\square$

Thm 5.3.1

Let X: rand var p.m.f $p(x)$

p.m.f $q(x)$

$\Rightarrow$ error. $\Rightarrow D(p\|q)$

Thm 5.4.3. The expected length under $p(x)$ of the code length $l(x) = \left\lceil \log \frac{1}{q(x)} \right\rceil$

$$H(p) + D(p\|q) \le E_p l(x)$$

$$\le H(p) + D(p\|q) + 1$$

Pf "$\le$". $E\, l(x) = \sum p(x) \left\lceil \log \frac{1}{q(x)} \right\rceil$

$$\le \sum p(x) \left(\log \frac{p(x)}{q(x)} \frac{1}{p(x)} + 1 \right)$$

$$= D(p\|q) + H(p) + 1$$

"$\geq$" $$E\,l(x) > \sum_x p(x) \log \frac{1}{q(x)}$$

$$= \sum p(x) \log \frac{p(x)}{q(x)} \frac{1}{p(x)}$$

$$= D(p\|q) + H(p)$$

wrong dist q gives penalty

$$\underline{D(p\|q)}$$

§5.5 Kraft ineq for Unique decodable codes

Review

extension 1-1

§5.2 inst code $\Leftrightarrow$ Kraft $\sum D^{-l_i} \le 1$

§5.5 Uni decodably $\Leftrightarrow$ //

Unique decodable $\overset{\text{length}}{=}$ inst.

Thm 5.5.1 The codeword lengths of Uniquely decodable D-ary code must satisfy

$$\sum D^{-l_i} \le 1$$

Pf) For extension code

$$l(x_1 \dots x_n) = \sum_{i=1}^{n} l(x_i)$$

$$\left(\sum_{x\in X} D^{-l(x)}\right)^{k} = \underbrace{\sum_{x_1\in X}\cdots\sum_{x_n\in X}} D^{\overbrace{-l(x_1)-\dots-l(x_n)}^{k}}$$

1, 2, ..., $k \cdot l_{max}$

$$= \sum_{m=1}^{k l_{max}} \underbrace{a(m)}_{\le D^m} D^{-m}$$

$a(m)$: # of code word with length m

$a(m) \le D^m$

$$\left(\sum_{x \in X} D^{-l(x)}\right)^{k} \le k\, l_{max} \quad \frac{1}{k}$$

$$\sum_{x \in X} D^{-l(x)} \le \underline{(k\, l_{max})^{\frac{1}{k}}}$$

$$\to 1$$

as $k \to \infty$

$$\sum D^{-l_i} \le 1 \quad \boxed{KI}$$

Conclusion.

Uniq decodable $\overset{\text{length}}{=}$ Inst

§ 5.6 Huffman code

Inst.

Ex 5.6.1 For 2-ary alphabet

X	$\Pr(X=x)$	$C(x)$								
1	0.25	11			0.3	10	0.45	0	0.45	0
2	0.25	00			0.25	11	0.3	10	0.55	1
3	0.2	01			0.25	00	0.25	11		
4	0.15		0.3	100	0.2	01				
5	0.15			111						

0.45

not 2-ary $D \geq 3$

3-ary

Ex 5.6.3 D: 3-ary alphabet $1 + k(D-1)$

$1 + k \cdot 2$

X	$Pr(X=x)$	
1	0.25	1
2	0.25	2
3	0.2	01
4	0.1	02
5	0.1	000
6	0.1	001
Dummy	0	002

0.25	1
0.25	2
0.2	00
0.2	01
0.1	02

0.5	− 0
0.25	− 1
0.25	− 2